```
public class EightQueens {
   static int[] board = new int[8];
   static boolean safe(int row, int col) {
      for (int i = 0; i < row; i++) {
         if (board[i] == col ||
            Math.abs(board[i] - col) == Math.abs(i - row))
            return false;
      }
      return true;
   }
   static boolean solve(int row) {
      if (row == 8)
         return true;

      for (int col = 0; col < 8; col++) {
         if (safe(row, col)) {
            board[row] = col;

            if (solve(row + 1))
               return true;
         }
      }
      return false;
   }
   public static void main(String[] args) {
      if (solve(0)) {
```

```
    for (int i = 0; i < 8; i++) {
      for (int j = 0; j < 8; j++)
        System.out.print(board[i] == j ? "Q " : ". ");
      System.out.println();
    }}}}
```

1. Theory

Constraint Satisfaction Problem (CSP):

A CSP is a problem in which variables are assigned values while satisfying given constraints.

8-Queens Problem:

- Place 8 queens on an 8×8 chessboard.
- No two queens should attack each other.
- A queen attacks horizontally, vertically and diagonally.

CSP Representation:

- Variables: Q1 to Q8 (8 queens).
- Domain: Columns 1 to 8.
- Constraints: No two queens in the same column or diagonal.
- Only one queen is placed in each row.

Backtracking:

Backtracking is a technique that places queens one by one. If a position is unsafe, it tries another position. If no position is available, it goes back and changes the previous queen's position.

2. Algorithm

1. Start with an empty chessboard.
2. Begin from the first row.
3. Try each column.
4. Check whether the position is safe.

5. If safe, place the queen and move to the next row.
6. If unsafe, try another column.
7. If no column is safe, backtrack.
8. Repeat until all 8 queens are placed.
9. Display the solution.
10. Stop.

3. Example Solution

Solution array: [1, 5, 8, 6, 3, 7, 2, 4]

Q.....

....Q...

.....Q

.....Q..

..Q.....

.....Q.

.Q.....

...Q....

4. Viva Questions and Answers

Q1. What is CSP?

A problem where variables are assigned values while satisfying constraints.

Q2. What are the variables in 8-Queens?

Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8.

Q3. What is the domain?

Columns from 1 to 8.

Q4. What are the constraints?

No two queens in the same row, column or diagonal.

Q5. What is backtracking?

Trying a solution and going back if it fails.

Q6. Why is backtracking used?

To find a valid arrangement of 8 queens.

Q7. How do you check a safe position?

Check the same column and both diagonals.

Q8. What is the diagonal condition?

$|Column1-Column2| \neq |Row1-Row2|$

Q9. Why is recursion used?

To place queens row by row and backtrack when necessary.

Q10. Give applications of backtracking.

Sudoku, N-Queens and graph coloring.

Q11. What is the size of the board?

8 × 8.

Q12. How many queens are placed in each row?

Exactly one.